Where to next for hamstrings? A biomechanical and anatomical perspective

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In recent years, there has been rising interest and debate on the importance of addressing running mechanics for preventing running-related hamstring injury. It is a logical and intriguing avenue of research given the high prevalence of such injuries. In theory, lower-limb mechanics directly relate to hamstring musculotendinous demands when running—the relative positions of the hip and knee joints, and pelvic tilt, will determine the hamstring muscle–tendon unit (MTU) lengths. Notably, hamstring MTUs during running are subject to large strain and force during late swing. However, there is a paucity of research prospectively investigating the relationship between running mechanics, or hamstring MTU dynamics, and injury, likely because it is challenging to prospectively obtain large samples of data from injured athletes.

ASSESSING RUNNING MECHANICS

To better understand the relationship between running mechanics and hamstring injury, it can be useful to examine multiple case studies (or case series), the advantage being they offer constructive observation(s) towards novel concepts in ecologically valid contexts. Schuermans *et al*[1] found that four injured soccer players displayed greater anterior pelvic tilt during mid-swing and greater thoracic lateral flexion during the late swing phase of sprinting. Similarly, in a study of three rugby players[2] who subsequently sustained hamstring injuries, a tendency for greater thoracic lateral flexion during the late swing phase as well as greater knee power absorption and hip extension moments was observed. As noted earlier, given the attachment sites of the hamstring muscles, it is logical that the mechanics of the hip, knee and pelvis may relate to hamstring strain and injury risk. It is also likely that the observed mechanics are interrelated, as any increase in joint moment magnitude that increases the acceleration of a segment (eg, a larger hip extension moment increasing the acceleration of the thigh) may influence adjacent joints and other segments through intersegmental torques. This may be especially evident during the swing phase where muscle-generated torques are primarily opposing motion-dependent torques.[3] The influence of thoracic lateral flexion is perhaps less clear, but it has been postulated that this may relate to insufficient dissociative capacity within the lumbo–pelvic–hip complex.[1] Increased thoracic lateral flexion may also relate to the observed hip extension moment in injured athletes, given that during late swing, energy generated by the hip is transferred proximally to the trunk.[2] Collectively, these data from soccer and rugby show emerging evidence that running mechanics may influence hamstring injury risk. However, data are from a small number of case studies; therefore, although insightful, evidence remains inconclusive. Further, both studies[1 2] examined mechanics in a laboratory setting without the presence of fatigue. This is important to note given that many injuries occur in the latter half of game periods[4] (eg, halves or quarters) under fatigued conditions, which may alter running mechanics.[5] The relationship between fatigue-affected running mechanics and hamstring injury warrants further research. Nevertheless, these findings prompt us to consider how mechanics at the musculotendon level may influence injury susceptibility.

MUSCULOTENDON MECHANICS

Several studies have attempted to link hamstring MTU mechanics to injury. Hamstring MTU lengths and forces directly prior to injury have been determined from serendipitous measurement during running-based injury.[6 7] These studies showed that during late swing, the injured leg's biceps femoris long head (BFlh) reached a greater length and produced greater net hamstring MTU force compared with the uninjured leg. These data suggest that BFlh strain and force are key mechanisms for injury. However, the athlete(s) in those studies had sustained recent recurrent hamstring strains on the injured leg prior to testing; therefore, it may be argued the observed MTU dynamics may be a consequence of the prior injury. A more recent report from two different injury cases provides additional evidence linking hamstring MTU dynamics to injury.[8] In this study, data from the pre-season prior to injury showed that the subsequently injured BFlh of both athletes displayed the greatest strain of all hamstring MTUs during late swing. A strength of this study was that MTU force was normalised to physiological cross-sectional area (PCSA), as PCSA reflects the force-generating capacity of the muscle. It was seen that BFlh MTU force was equal to, or greater than, semimembranosus force during late swing. This suggests that the BFlh muscle was under considerable stress during late swing due to the large force in a muscle with a relatively small PCSA. To our knowledge, hamstring MTU force has not previously been considered in this way; however, this insight prompts us to contemplate how information about running mechanics and MTU dynamics could be enhanced with anatomical information.

HAMSTRING SHAPE

When considering anatomical information alongside running mechanics and MTU dynamics, it is important to have some appreciation of what anatomical research has been conducted. Early research focused largely on cadaveric measures of overall morphology such as PCSA, volume and MTU length. Since then, research has explored the importance of muscle architecture, in particular fascicle length.[9] This has had considerable practical implications for injury prevention programmes, reinforcing the importance of eccentric hamstring exercises or exercises that stress the muscle at long lengths, to elicit fascicle elongation. More recently, advanced methods for capturing information about muscle morphology, including complex three-dimensional shape, have been used. These methods have the potential to provide much more information about hamstring morphology: how it varies between groups and how it may relate to injury risk. Sutherland *et al*[10] applied statistical shape modelling techniques to

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discriminate hamstring muscle shapes between rugby and sprinting athletes. Three-dimensional shape features such as size, curvature and torsion were found to distinguish the cohorts. This suggests that meaningful variation in hamstring shape exists between individuals and populations and can be captured in a way that could be used for understanding injury risk. To date, no studies have investigated whether three-dimensional muscle shape may relate to injury risk. However, this is an interesting avenue for future research, given the theoretical link between muscle shape and function. Variations in shape theoretically reflect a variation in the muscle's line of action,[11] which could influence MTU forces and forces transferred to the bone to produce joint movement. Given that biomechanical research has suggested that MTU forces may influence injury susceptibility,[8] it is plausible that muscle shape could provide additional insight into hamstring muscle function and could be linked to injury. Further, information about muscle shape could be used within musculoskeletal models to further integrate analyses of biomechanics and anatomy in relation to injury.

Moving forward, to gain a better understanding of hamstring function and mechanisms for injury, we should consider joint mechanics along with hamstring MTU dynamics and anatomy rather than exploring each in isolation.

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